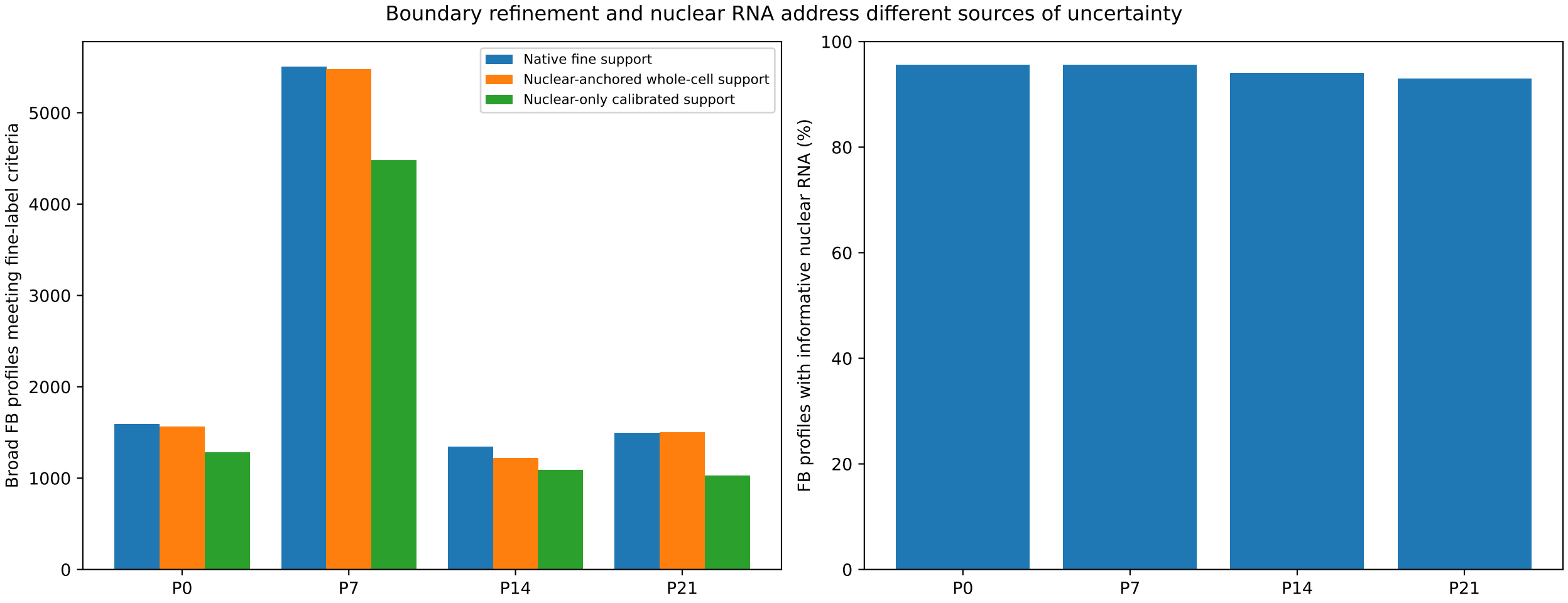
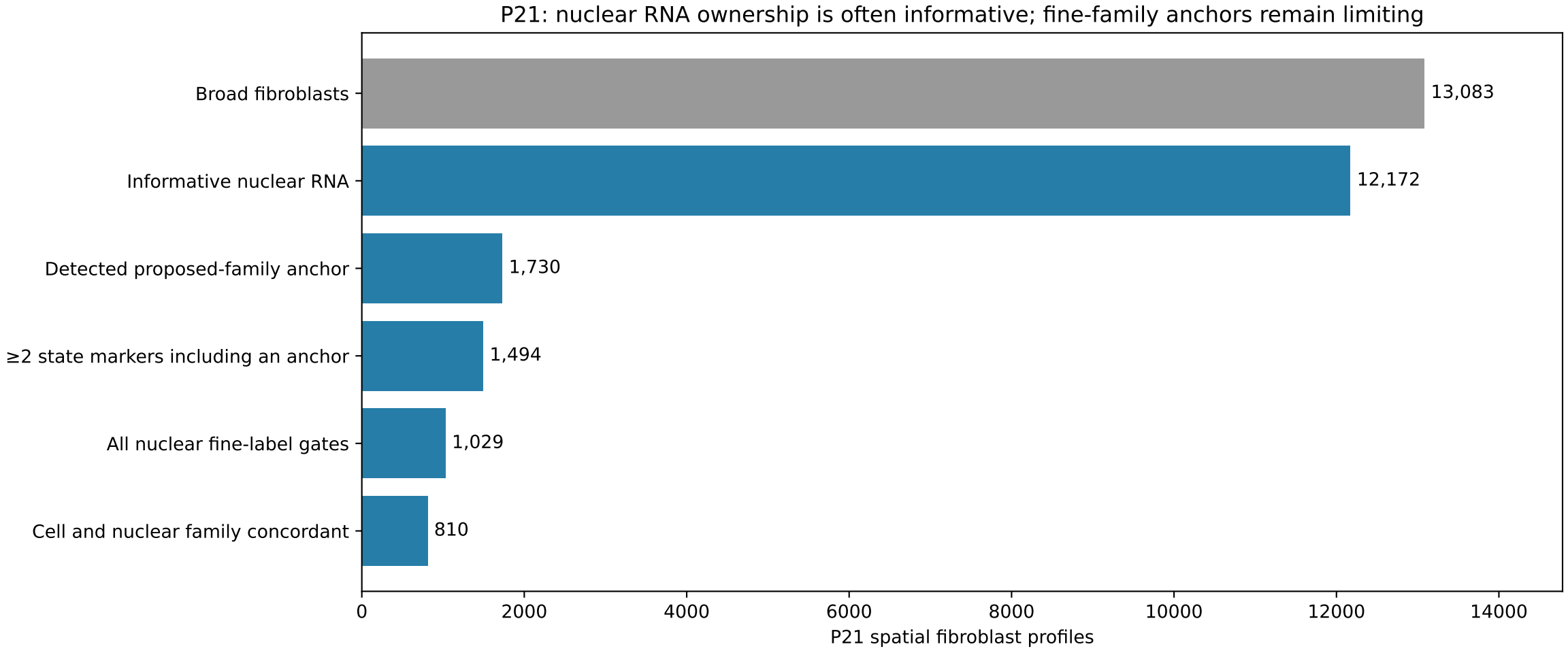
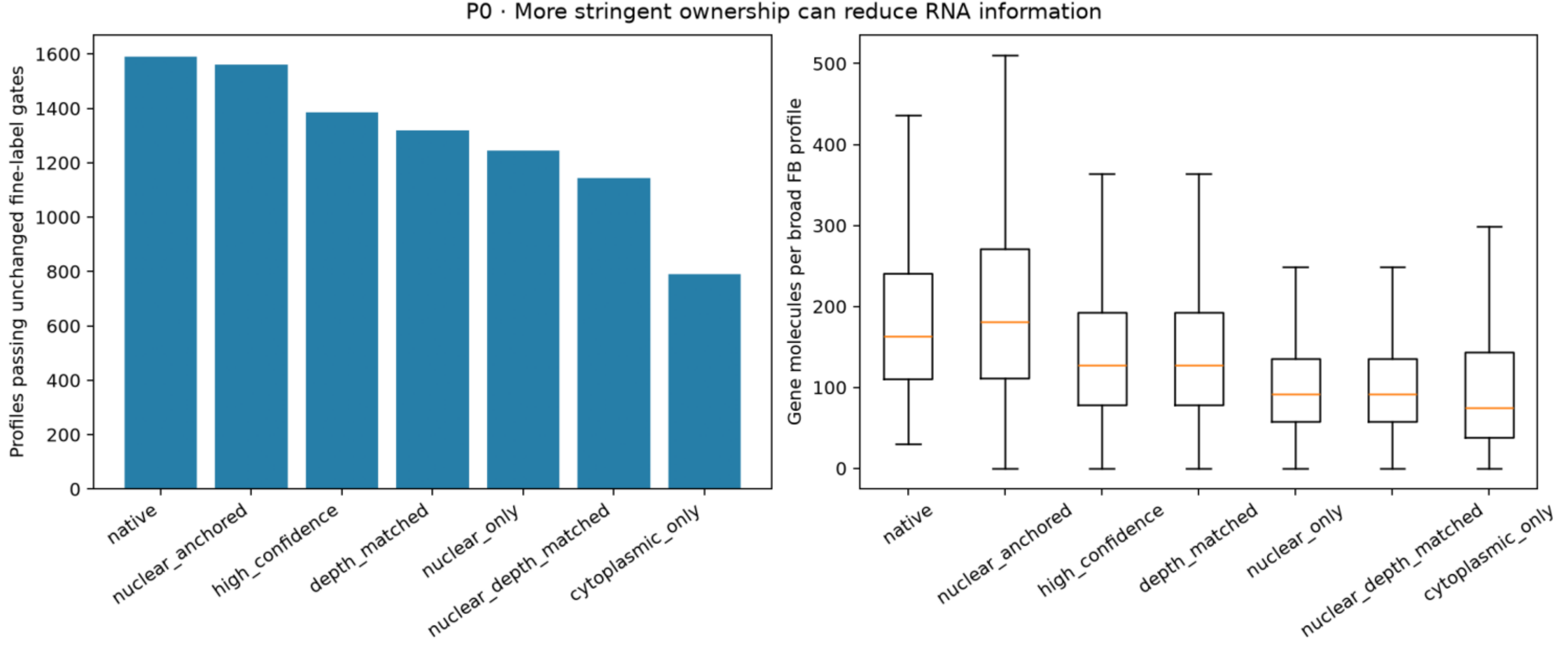
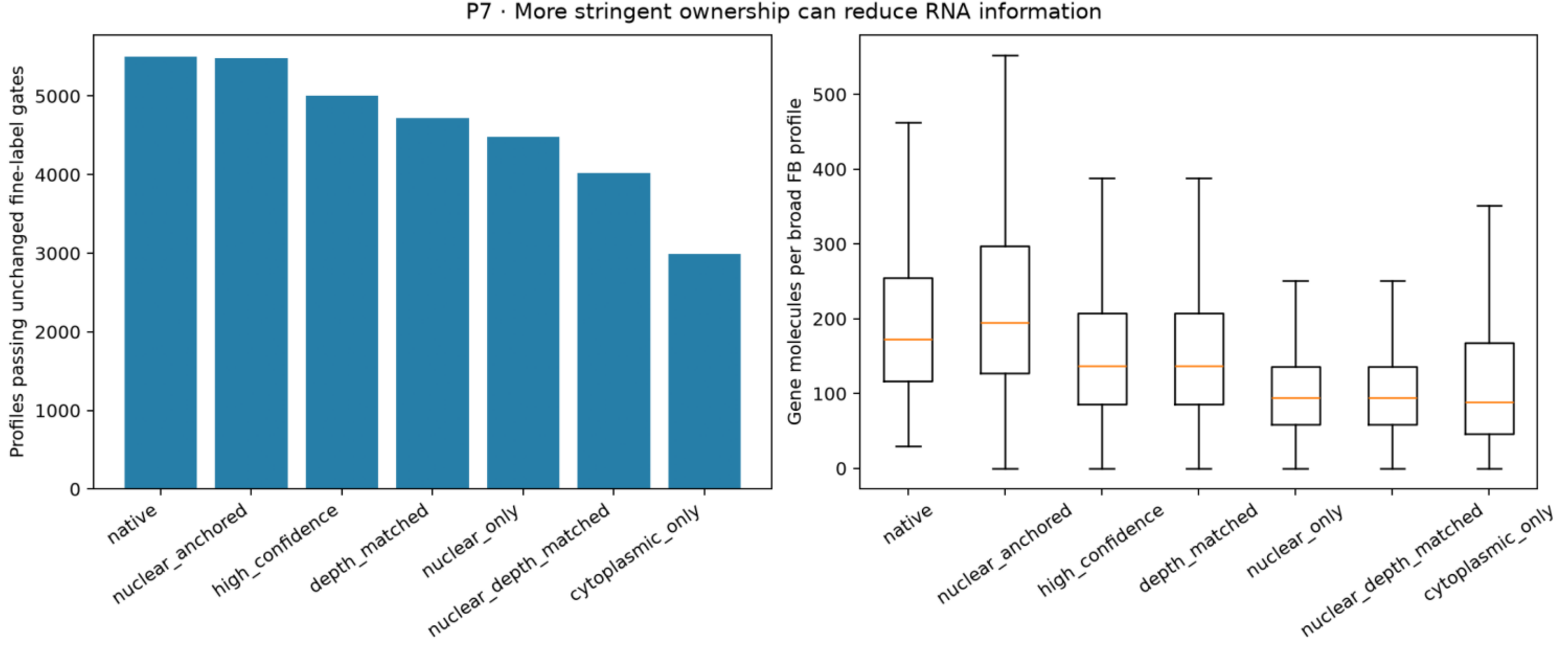

Boundary refinement and nuclear RNA address different sources of uncertainty
Native fine support
Nuclear-anchored whole-cell support
Nuclear-only calibrated support
Broad FB profiles meeting fine-label criteria
0
1000
2000
3000
4000
5000
P0
P7
P14
P21
FB profiles with informative nuclear RNA (%)
0
20
40
60
80
100
P0
P7
P14
P21

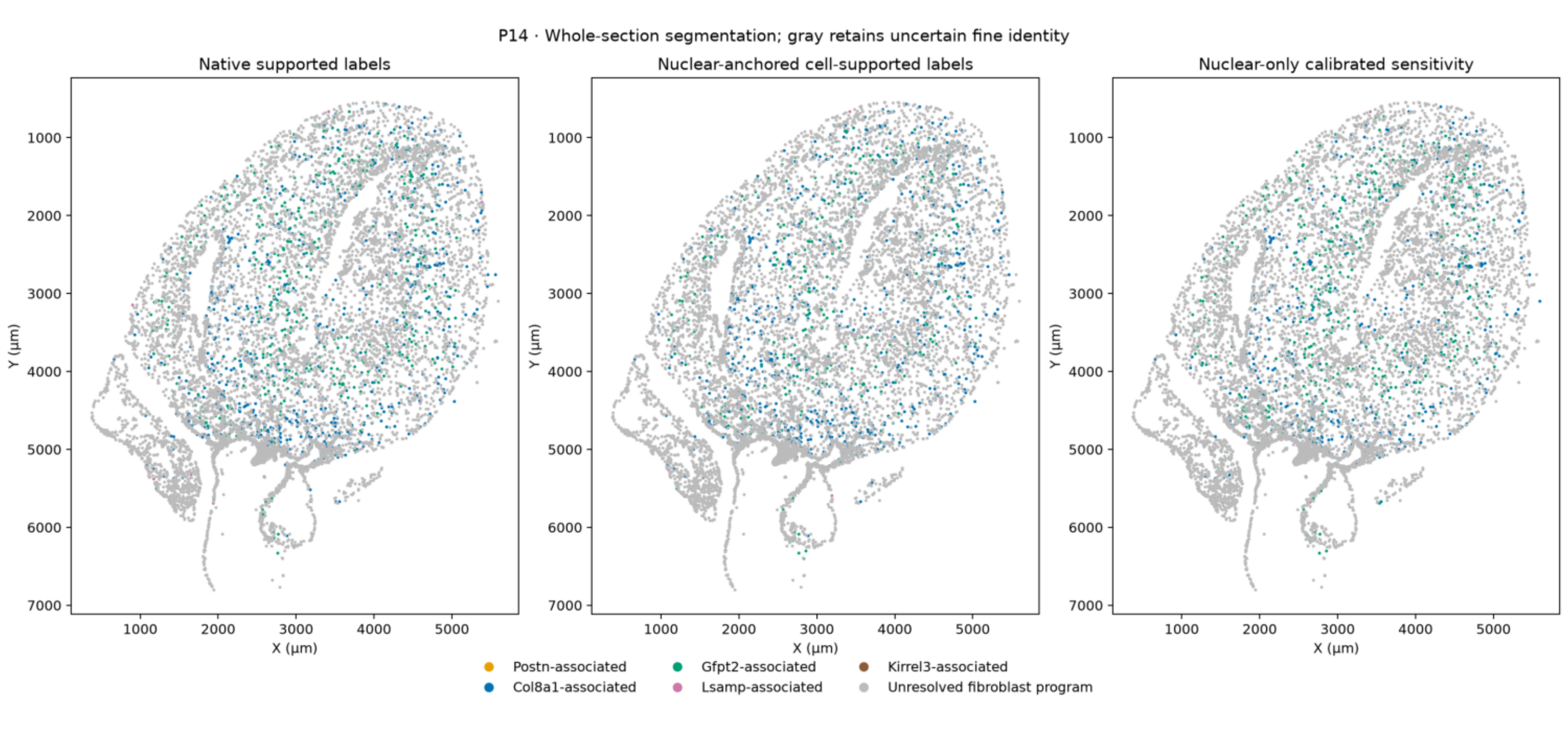

P21: nuclear RNA ownership is often informative; fine-family anchors remain limiting
Broad fibroblasts
Informative nuclear RNA
Detected proposed-family anchor
≥2 state markers including an anchor
All nuclear fine-label gates
Cell and nuclear family concordant
13,083
12,172
1,730
1,494
1,029
810
0
2000
4000
6000
8000
10000
12000
14000
P21 spatial fibroblast profiles

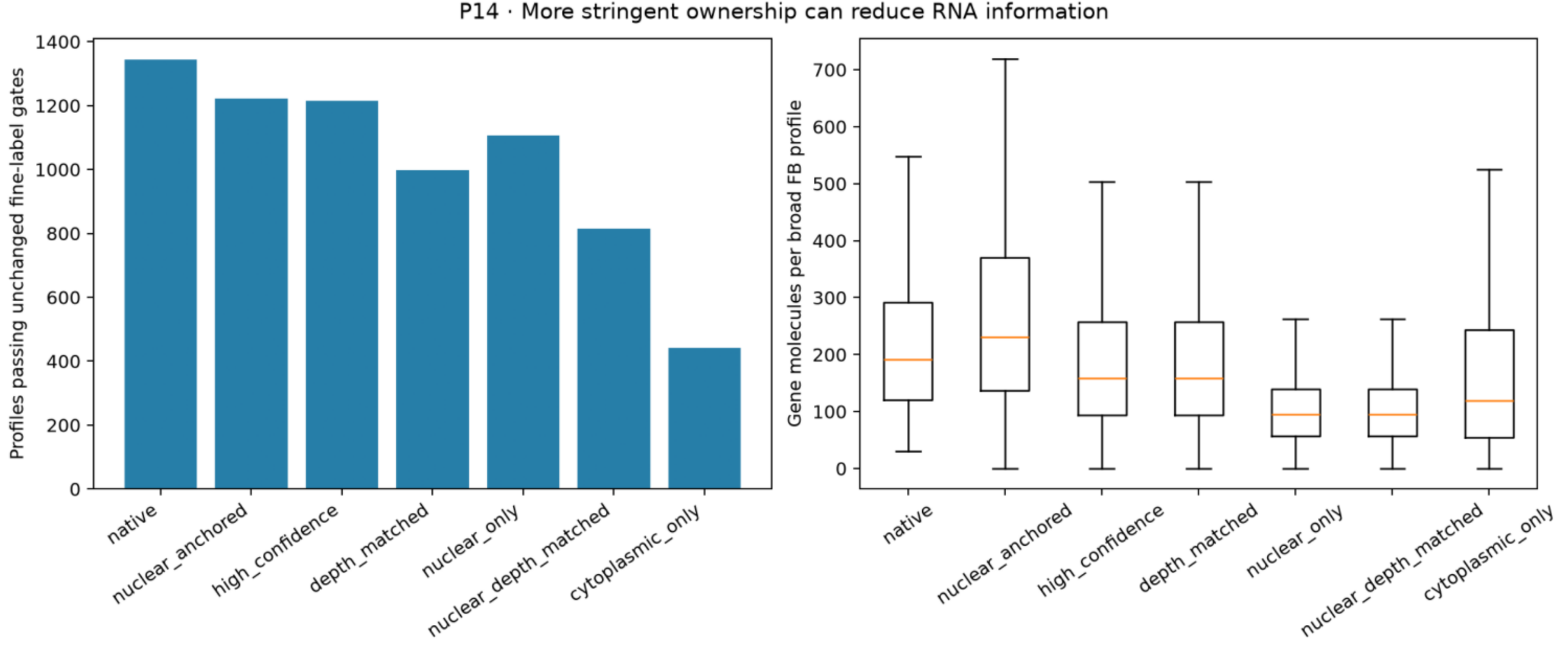

P0 · Whole-section segmentation; gray retains uncertain fine identity
Native supported labels
Nuclear-anchored cell-supported labels
Nuclear-only calibrated sensitivity
Y (µm)
X (µm)
500
1000
1500
2000
2500
3000
3500
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program

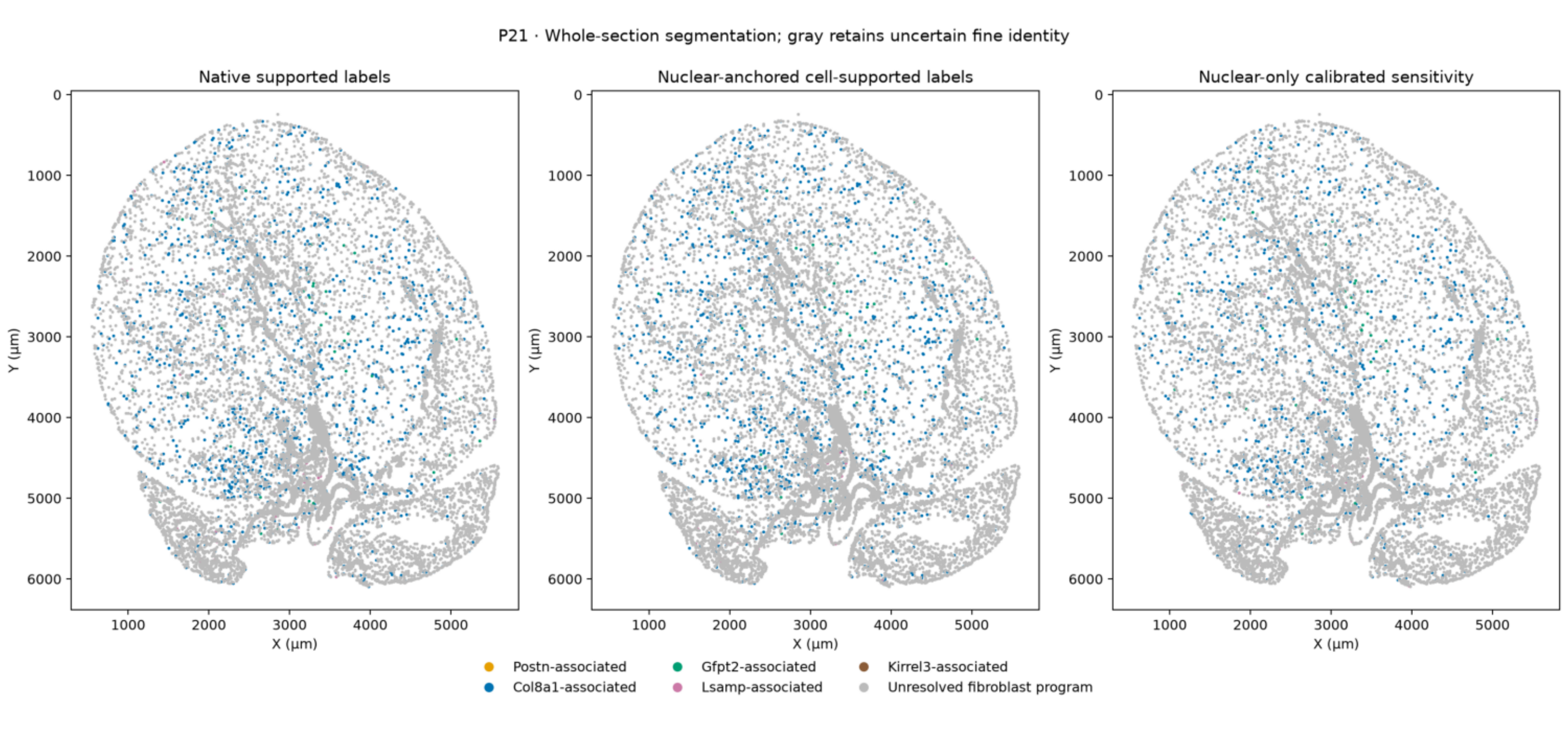

P0 · More stringent ownership can reduce RNA information
Profiles passing unchanged fine-label gates
0
200
400
600
800
1000
1200
1400
1600
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only
Gene molecules per broad FB profile
0
100
200
300
400
500
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only

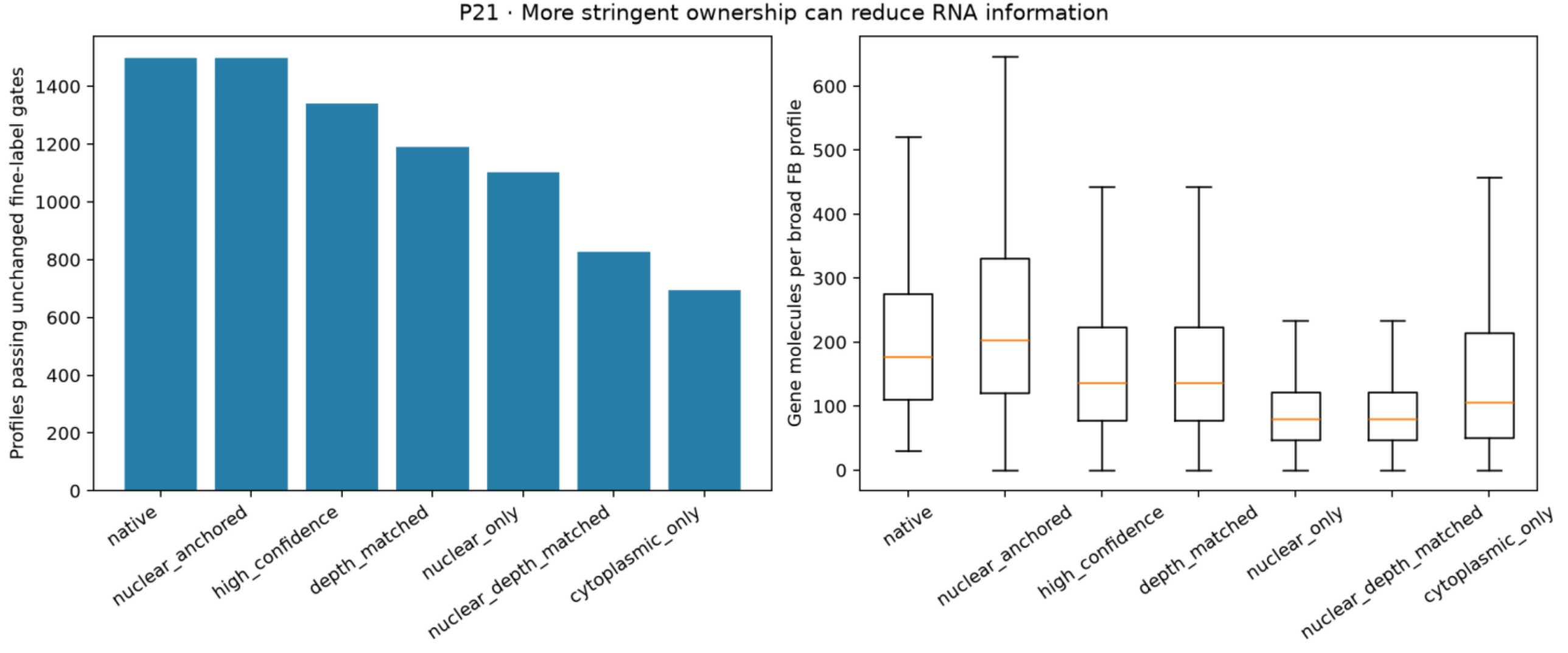

P7 · Whole-section segmentation; gray retains uncertain fine identity
Native supported labels
Nuclear-anchored cell-supported labels
Nuclear-only calibrated sensitivity
Y (µm)
X (µm)
500
1000
1500
2000
2500
3000
3500
4000
4500
1000
2000
3000
4000
5000
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program


P7 · More stringent ownership can reduce RNA information
Profiles passing unchanged fine-label gates
0
1000
2000
3000
4000
5000
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only
Gene molecules per broad FB profile
0
100
200
300
400
500
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only


P14 · Whole-section segmentation; gray retains uncertain fine identity
Native supported labels
Nuclear-anchored cell-supported labels
Nuclear-only calibrated sensitivity
Y (µm)
X (µm)
1000
2000
3000
4000
5000
6000
7000
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program


P14 · More stringent ownership can reduce RNA information
Profiles passing unchanged fine-label gates
0
200
400
600
800
1000
1200
1400
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only
Gene molecules per broad FB profile
0
100
200
300
400
500
600
700
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only


P21 · Whole-section segmentation; gray retains uncertain fine identity
Native supported labels
Nuclear-anchored cell-supported labels
Nuclear-only calibrated sensitivity
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
Postn-associated
Col8a1-associated
Gfpt2-associated
Lsamp-associated
Kirrel3-associated
Unresolved fibroblast program


P21 · More stringent ownership can reduce RNA information
Profiles passing unchanged fine-label gates
0
200
400
600
800
1000
1200
1400
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only
Gene molecules per broad FB profile
0
100
200
300
400
500
600
native
nuclear_anchored
high_confidence
depth_matched
nuclear_only
nuclear_depth_matched
cytoplasmic_only